Water Conservation and Rainwater Harvesting

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Introduction to Water Conservation

Water conservation means using water wisely, reducing waste and protecting water sources so that enough clean water remains for present and future generations. Although about 71 percent of the Earth is covered with water, only about 3 percent is fresh water, and most of that is frozen in glaciers and ice caps, so less than 1 percent is easily available in rivers, lakes and groundwater. India supports about 18 percent of the world's population but has only around 4 percent of the world's freshwater resources. Per-person water availability in India has fallen to around 1,400 to 1,500 cubic metres per year, below the 1,700 cubic metre level regarded as water-stressed. NITI Aayog has warned that many Indian cities face serious shortages. World Water Day is observed on 22 March every year.

What Is Rainwater Harvesting?

Rainwater harvesting is the collection and storage of rainwater from rooftops, land surfaces or other catchments so that it can be used later or allowed to soak into the ground to recharge groundwater. Instead of letting rain run off into drains and the sea, the water is captured where it falls. There are two main approaches. In the first, rainwater is stored in tanks, sumps or ponds for direct use, such as for washing, gardening, toilets and, after treatment, drinking. In the second, rainwater is directed into the ground through recharge pits, trenches or recharge wells so that the groundwater level rises and borewells and wells yield more water.

How Rooftop Rainwater Harvesting Works

A typical rooftop system has a catchment area (the roof), gutters and downpipes to carry the water, a filter or first-flush device that discards the dirty first rain, and a storage tank or a recharge structure. Rainwater from the roof is led to a filter chamber containing sand, gravel and charcoal, and the clean water either goes into an underground tank or flows into a recharge well or pit that connects to the aquifer. The system is cheap and simple, can be installed in houses, apartments, schools and offices, and the water is naturally soft and free from salts. Other structures include percolation tanks, check dams, contour trenches, farm ponds and the restoration of lakes and traditional tanks. Large-scale watershed management applies similar ideas across entire catchments.

Rainwater Harvesting in Tamil Nadu

Tamil Nadu has a long tradition of rainwater storage through eris (tanks), temple ponds called ooranis and a community maintenance practice called kudimaramathu, with about 40,000 tanks in the state. Modern rainwater harvesting began after severe groundwater depletion and drought in Chennai. In 2003, Tamil Nadu became the first state in India to make rainwater harvesting mandatory for all buildings, through an ordinance issued in July 2003 and then legislated as an amendment to the municipal laws, under Chief Minister J. Jayalalithaa. All new and existing buildings, whether residential, commercial or government, had to install rainwater harvesting structures within a short deadline. The campaign included a large public awareness drive, and rainwater structures were built in lakhs of buildings. Studies after 2003 noted a rise in groundwater levels in Chennai and improved water quality in many areas. Tamil Nadu's example was later followed by other states and cities. The government also began Kudimaramathu schemes for repairing tanks and has promoted the restoration of ponds, and Chennai's Pallikaranai marsh and numerous lakes are important to flood and water management. However, enforcement over the years has been uneven, and the Chennai city was hit by a severe water crisis in the summer of 2019, when major reservoirs ran almost dry, which renewed the push for harvesting and recycling.

Other Government Initiatives

The Jal Shakti Abhiyan, launched in 2019 with the slogan "Catch the Rain, Where it Falls, When it Falls", promotes conservation and rainwater harvesting in water-stressed districts. The Atal Bhujal Yojana, started in 2019, supports community-led groundwater management in seven states. The Jal Jeevan Mission supplies tap water to rural homes, and the Amrit Sarovar mission (2022) aims to build or rejuvenate 75 water bodies in every district. In Rajasthan, Rajendra Singh, known as the Waterman of India, revived rivers through the traditional johad check dams and won the Stockholm Water Prize in 2015.

Three Ways to Save Water at Home

Three simple ways to save water at home are to fix leaking taps and pipes promptly, to turn off the tap while brushing teeth or washing hands and use a bucket rather than a running shower or hose, and to reuse water, such as using the water from washing vegetables or the reject water from an RO purifier to mop floors or water plants. Other methods include running washing machines and dishwashers only with full loads, installing dual-flush toilets and low-flow taps, watering plants early in the morning or in the evening with drip irrigation, washing vehicles with a bucket rather than a hose, collecting rainwater in a barrel, and spreading mulch around plants. A dripping tap can waste thousands of litres in a year. Teaching children to value water is also important.

Water Conservation in Agriculture and Industry

Agriculture uses about 80 percent of India's fresh water, so efficient irrigation is vital. Drip and sprinkler irrigation can save 30 to 50 percent of water compared with flooding fields. Crop choice also matters, as rice and sugarcane use large amounts of water, and millets use much less. Industries can adopt recycling and Zero Liquid Discharge.

Challenges and Future

Challenges include over-extraction of groundwater, pollution, climate-driven erratic rainfall, urban concrete surfaces that prevent soaking and weak enforcement. Future approaches include sponge city designs, smart water meters, recycling of treated wastewater and community participation.

Quick Facts

- Rainwater harvesting is collecting and storing rain or recharging groundwater with it.
- Tamil Nadu made rainwater harvesting mandatory in 2003, the first state in India to do so.
- Three home water-saving tips: fix leaks, turn off taps when not needed, and reuse water.
- India has about 18 percent of the world's people but only about 4 percent of its fresh water.
- World Water Day is observed on 22 March.